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# LAB REPORT II

## (INSERTION SORTING)

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**Machine Problem Description: Faulty Machine Finder in a Factory**

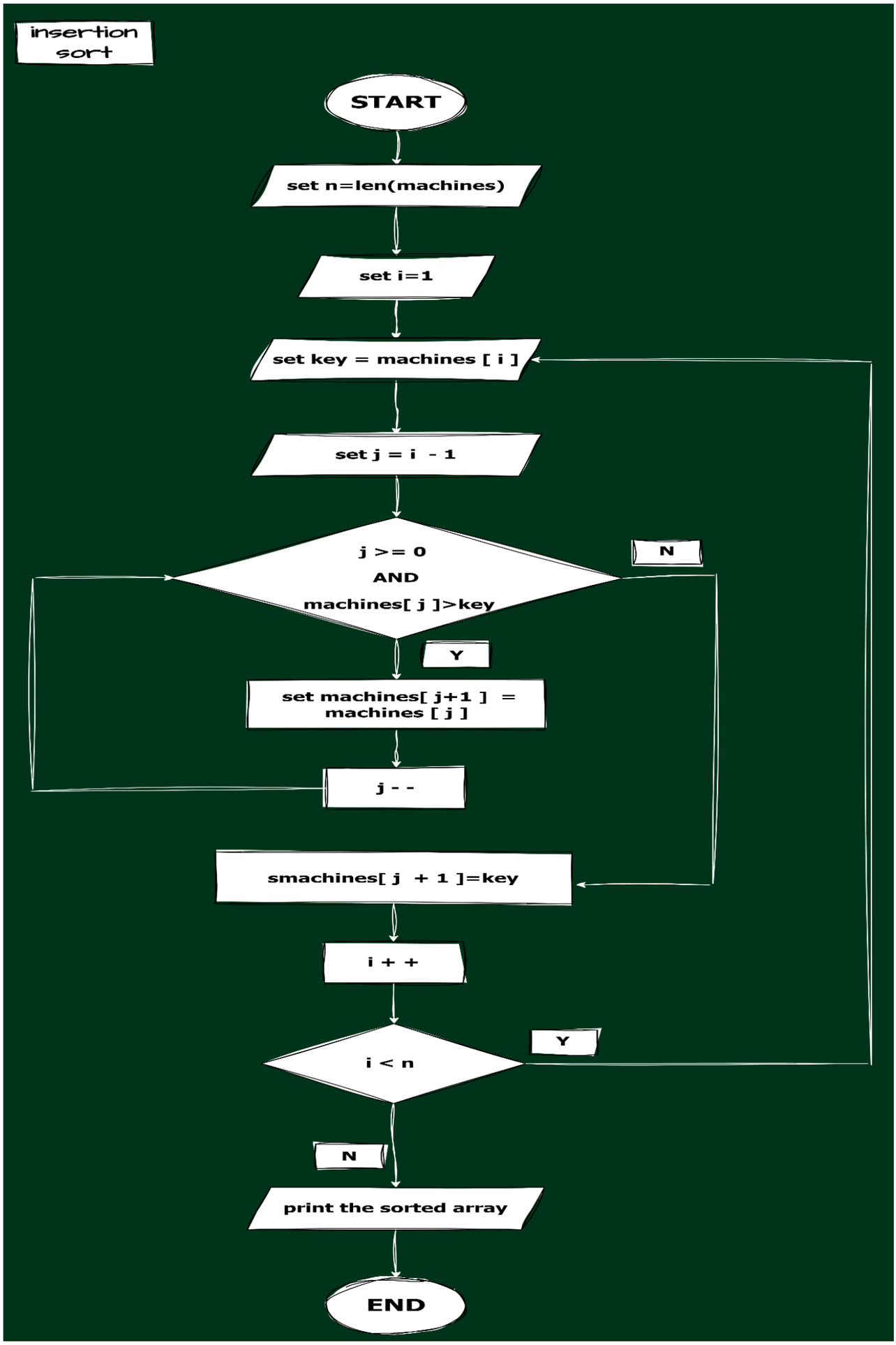
In a factory, multiple machines are producing items. Each machine is identified by a unique ID, and most machines produce an even number of items, meaning their IDs appear twice in the list of machines. However, one machine is faulty and produces an odd number of items, which means that the faulty machine's ID appears only once in the list. The goal is to sort the list of machine IDs using different sorting algorithms and then identify the faulty machine by applying the **XOR operation** to find the ID that appears only once.

## INSERTION SORT:

This algorithm sorts the list constructing an ordered part from one element at a time. It repeatedly takes the next unsorted element and inserts it in its correct position in the ordered part.

# I FLOWCHART (INSERTION SORT):

In the following flowchart, the program uses Insertion Sort on a single sequence of machine IDs.



# SOURCE CODE:

In the following code, the program uses Insertion Sort on a single sequence of machine IDs.

FaultyMachine_InsertionSorting.py

```python
#perfor insertion sort in this function
def insertionSort(machines):
    n = len(machines)
    for i in range(1, n):
        key = machines[i]
        j = i - 1
        while j >= 0 and machines[j] > key:
            machines[j + 1] = machines[j]
            j -= 1
        machines[j + 1] = key
    return machines
def findFaultyMachine(machines):
    result = 0
    for machine in machines:
        result ^= machine
    return result
def main():
    machines = [303, 101, 404, 202, 101, 303, 404]
    print(f"Original machines: {machines}")
    sorted_machines = insertionSort(machines)
    print(f"Sorted machines using Insertion Sort: {sorted_machines}")
    faultyMachine = findFaultyMachine(sorted_machines)
    print(f"The faulty machine is: {faultyMachine}")
if __name__ == "__main__":
    main()
```

## OUTPUT:

In this program, the machine IDs are sorted in ascending order, then printed as an array.

```
"C:\Users\Gian Carlo Trilles\PycharmProjects\pythonProject\.venv\Scripts\python.exe" "C:\
Original machines: [303, 101, 404, 202, 101, 303, 404]
Sorted machines using Insertion Sort: [101, 101, 202, 303, 303, 404, 404]
The faulty machine is: 202

Process finished with exit code 0
```